PEPII

RF SYSTEM

RF STATION COUPLING & CABLE CAL PROCEDURE

PS-340-330-56-R0

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Station #:_____ Date:_____ Initl.:_____

RF STATION COUPLING & CABLE LOSS TEST PROCEDURE & DATA

1) Introduction

In the attached table data is to be entered derived from component calibration data sheets or the cable loss is to be calibrated using the procedure described below.

2) Procedure

Cavity sampling probe coupling value:
In the column "Device Info" enter the Cavity Raft Assembly # installed in cavity position A, B, C or D.
Cavity sampling probe database value is nominal 99.6 dB modified by a number derived from the LOW POWER TEST DATA for the correct Cavity Raft Assembly. On page 3 LOW POWER TEST DATA calculate the difference between the calculated coupling P_s/P_{inc} (approx. -52.5 dB) and the "Final loop coupling number after tightening". If the final number is larger than P_s/P_{inc} the difference must be added to the 99.6 dB so it will also be larger.

Forward and reflected coupling value:
In the column "Device Info" enter the Coupler # installed and check if the FORWARD port of the coupler agrees with the forward direction of the power flow. If the coupler is reversed, please note.
The forward and reflected coupling value is found on calibration data sheets for the couplers.

Cable loss value:
The cable loss value needs to be measured at 476 MHz. First set the signal generator with a 10 dB matching attenuator to 0 dBm (1 mW) into the HP 435 power, then connect the signal generator/attenuator to the proper J-NUMBER of the 1/4 inch Heliax pigtails in the Blue Rack and measure the loss at the other end of the cable in the tunnel or upstairs with the same HP 435 power meter. Re-check the output power of the signal generator (0 dBm) often.

3) Equipment:

Signal Generator HP 8648

10 dB attenuator to improve source match

Power Meter HP 435

Station #:_____

Signal Name	Nominal dB	Measured dB
cavity A probe coupling value - cavity assembly # =	99.6	
cavity A probe Heliax, coupler and SMA pigtail cable loss (J1)	1.0	
cavity A probe IQ module channel conversion loss	13.15	
cavity A forward coupling value - coupler # =	60.0	
cavity A forward Heliax and SMA pigtail cable loss (J2)	1.0	
cavity A forward IQ module channel conversion loss	13.15	
cavity A reflected coupling value - coupler # =	60.0	
cavity A reflected Heliax and SMA pigtail cable loss (J3)	1.0	
cavity A reflected IQ module channel conversion loss	13.15	
cavity B probe coupling value and cavity assembly number =	99.6	
cavity B probe Heliax, coupler and SMA pigtail cable loss (J4)	1.0	
cavity B probe IQ module channel conversion loss	13.15	
cavity B forward coupling value - coupler # =	60.0	
cavity B forward Heliax and SMA pigtail cable loss (J5)	1.0	
cavity B forward IQ module channel conversion loss	13.15	
cavity B reflected coupling value - coupler # =	60.0	
cavity B reflected Heliax, and SMA pigtail cable loss (J6)	1.0	
cavity B reflected IQ module channel conversion loss	13.15	
cavity C probe coupling value and cavity assembly number =	99.6	
cavity C probe Heliax, coupler and SMA pigtail cable loss (J7)	1.0	
cavity C probe IQ module channel conversion loss	13.15	
cavity C forward coupling value - coupler # =	60.0	
cavity C forward Heliax and SMA pigtail cable loss (J8)	1.0	
cavity C forward IQ module channel conversion loss	13.15	
cavity C reflected coupling value - coupler # =	60.0	
cavity C reflected Heliax and SMA pigtail cable loss (J9)	1.0	
cavity C reflected IQ module channel conversion loss	13.15	
cavity D probe coupling value and cavity assembly number =	99.6	
cavity D probe Heliax, coupler and SMA pigtail cable loss (J10)	1.0	
cavity D probe IQ module channel conversion loss	13.15	
cavity D forward coupling value - coupler # =	60.0	
cavity D forward Heliax, and SMA pigtail cable loss (J11)	1.0	
cavity D forward IQ module channel conversion loss	13.15	
cavity D reflected coupling value - coupler # =	60.0	
cavity D reflected Heliax and SMA pigtail cable loss (J12)	1.0	

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cavity D reflected IQ module channel conversion loss	13.15	
klystron forward coupling value - coupler # =	60.0	
klystron forward Heliax and SMA pigtail cable loss (J15)	5.0	
klystron forward IQ module channel conversion loss	13.15	
klystron reflected coupling value - coupler # =	60.0	
klystron reflected Heliax and SMA pigtail cable loss (J16)	1.0	
klystron reflected IQ module channel conversion loss	13.15	
circulator load forward coupling value - coupler # =	60.0	
circulator load forward Heliax and SMA pigtail cable loss (J19)	4.0	
circulator load forward IQ module channel conversion loss	13.15	
circulator load reflected coupling value - coupler # =	60.0	
circulator load reflected Heliax and SMA pigtail cable loss (J20)	4.0	
circulator load reflected IQ module channel conversion loss	13.15	
magic tee #1 load forward coupling value - coupler # =	60.0	
magic tee #1 load forward Heliax and SMA pigtail cable loss (J21)	4.0	
magic tee #1 load forward IQ module channel conversion loss	13.15	
magic tee #1 load reflected coupling value - coupler # =	60.0	
magic tee #1 load reflected Heliax and SMA pigtail cable loss (J22)	4.0	
magic tee #1 load reflected IQ module channel conversion loss	13.15	
magic tee #2 load forward coupling value - coupler # =	60.0	
magic tee #2 load forward Heliax and SMA pigtail cable loss (J23)	4.0	
magic tee #2 load forward IQ module channel conversion loss	13.15	
magic tee #2 load reflected coupling value - coupler # =	60.0	
magic tee #2 load reflected Heliax and SMA pigtail cable loss (J24)	4.0	
magic tee #2 load reflected IQ module channel conversion loss	13.15	
magic tee #3 load forward coupling value - coupler # =	60.0	
magic tee #3 load forward Heliax and SMA pigtail cable loss (J25)	1.0	
magic tee #3 load forward IQ module channel conversion loss	13.15	
magic tee #3 load reflected coupling value - coupler # =	60.0	
magic tee #3 load reflected Heliax and SMA pigtail cable loss (J26)	1.0	
magic tee #3 load reflected IQ module channel conversion loss	13.15	
klystron drive forward coupling value - amplifier serial # =	30.0	
klystron drive forward Heliax and SMA pigtail cable loss (J36)	1.0	
klystron drive forward IQ module channel conversion loss	13.15	
klystron drive amplifier to klystron input forward cable loss (J14)	1.0	